

SymPy Bug Card

Bug 6: left limit gives the right-side value

Task. Evaluate $\lim_{x \rightarrow 0^-} p(x)$ of the discontinuous function

$$p(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

SymPy's answer. 1.

Correct answer. 0.

Explanation. For every sufficiently small negative x , the condition $x < 0$ is true, so the Piecewise expression is identically 0 on the left side of the jump. SymPy returns 1, the value of the right branch.

Likely root cause. The diagnosis points to `Piecewise._eval_as_leading_term` in `sympy/functions/elementary/piecewise.py`: it receives the approach direction `cdir` but substitutes the boundary value into branch conditions instead of using the requested side. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-006-left-limit-gives-the-right-side-value/](#)